SCS 1311 – Internet and Web Technologies

HTML

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Table Caption

- **<caption>** gives a title to the table
- Improves clarity and accessibility
- Appears above the table by default

```
<table>
  <caption>Student Marks</caption>
  ...
</table>
```

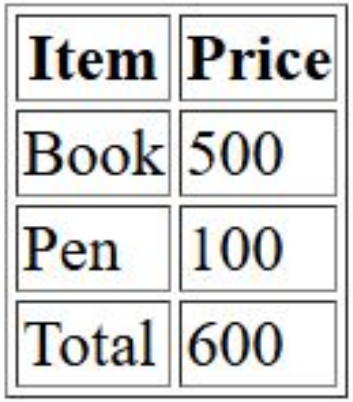
Student Marks

Name	Marks
Nimal	18
Kamal	20

Table Sections

- HTML provides table grouping elements:
  - <thead> → table header section
  - <tbody> → table body
  - <tfoot> → table footer
- Used for,
  - Better structure
  - Accessibility

Table Sections Cont.



Item	Price
Book	500
Pen	100
Total	600

```
<thead>
  <tr>
    <th>Item</th>
    <th>Price</th>
  </tr>
</thead>
<tbody>
  <tr>
    <td>Book</td>
    <td>500</td>
  </tr>
  <tr>
    <td>Pen</td>
    <td>100</td>
  </tr>
</tbody>
<tfoot>
  <tr>
    <td>Total</td>
    <td>600</td>
  </tr>
</tfoot>
```

Merging Columns and Rows

- **colspan** merges cells horizontally
- Attribute applied to <td> or <th>
  - `<td colspan="2">Total</td>`

- **rowspan** merges cells vertically
  - `<td rowspan="2">Semester 1</td>`

Merging Columns and Rows

```
<tr>
  <th rowspan="2">Name</th>
  <th colspan="2">Marks</th>
</tr>
<tr>
  <th>Maths</th>
  <th>Science</th>
</tr>
<tr>
  <td>Nimal</td>
  <td>85</td>
  <td>90</td>
</tr>
```

Name	Marks	
	Maths	Science
Nimal	85	90

Images

- Images make web pages:
  - More engaging
  - Easier to understand
  - Visually informative
- Images are added using the <img> tag
- <img> is a void element

Images Cont.

- **src** attribute specifies the image source
- Can be,
  - Relative path

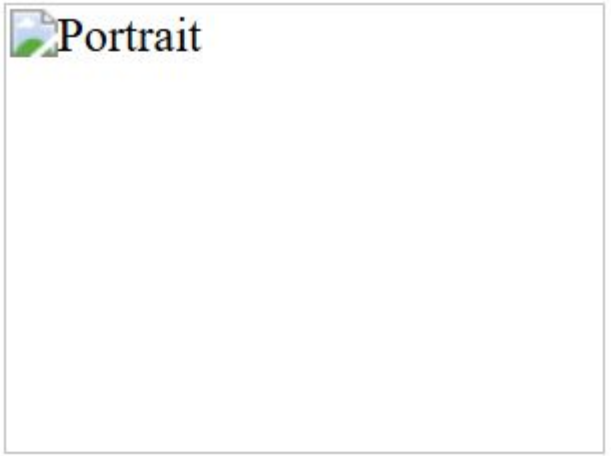
```
<img src="images/photo.jpg" alt="Student photo">
```

  - Absolute URL

```
<img src="https://example.com/logo.png" alt="Company logo">
```

Images Cont.

- alt provides alternative text
- Displayed when,
  - Image fails to load
  - Screen readers are used




```
<img src="photo.jpg" alt="Portrait">
```

Image Size and Dimensions

- Images have natural width and height
- Can set size using width and height attributes

```
<img src="photo.jpg" alt="Portrait"
width="200" height="150">
```

Images as Links

- Images can be clickable
- Wrap <img> inside an <a> tag

```
<a href="home.html">
  <img src="home.png" alt="Home page icon">
</a>
```

<figure> and <figcaption>

- Used to group images with captions
- Semantic and modern
- Use <figure> when,
  - Image needs a caption
  - Image is referenced in text
  - Image is independent from main text flow
  - Do not use <figure> for decorative images.

<figure> and <figcaption>

```
<figure>
  <img src="sunset.jpg" alt="Sunset over the beach">
  <figcaption>Sunset at the west coast</figcaption>
</figure>
```



Hyperlinks

What is a hyperlink?

- A hyperlink connects one web page to another
- Allows users to navigate between pages
- Created using the `<a>` (anchor) tag
- One of the core features of the web

Hyperlinks - Internal Links

```
<a href="about.html">About Us</a>
```

- <a> → anchor tag
- href → destination URL
- Links to another page in the same website
- Uses a relative path
- Common in navigation menus

Hyperlinks - External Links

```
<a href="https://www.ucsc.cmb.ac.lk">
    UCSC Website
</a>
```

- Links to another website
- Uses a full (absolute) URL
- Opens the target website in the browser

Hyperlinks vs File Paths

- A **hyperlink** defines **navigation**
- A **file path** defines **location**
- Hyperlinks **use** file paths

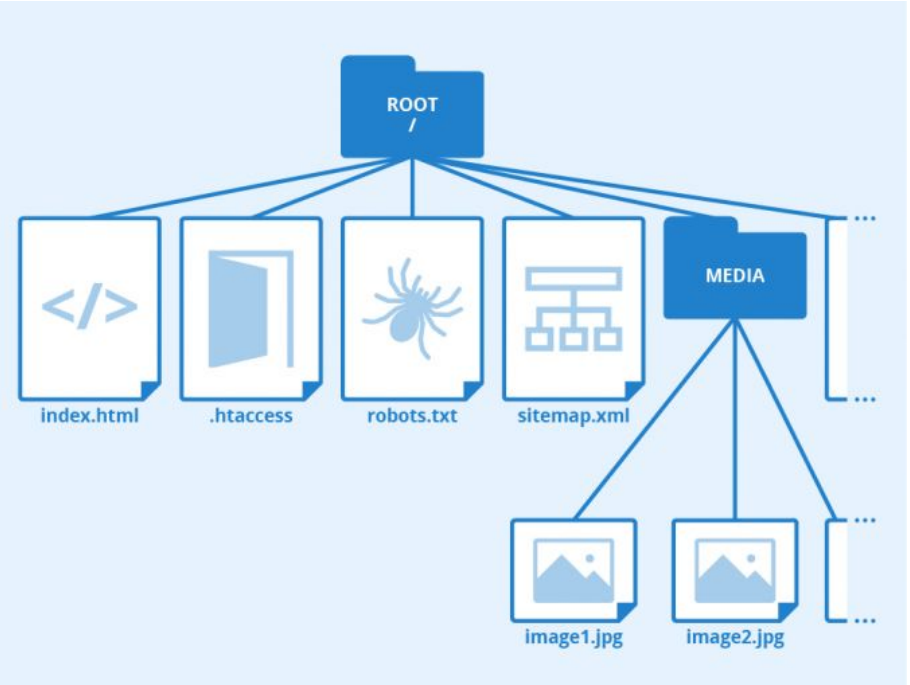
```
<a href="pages/contact.html">
<img src="images/logo.png">
<form action="submit.php">
```

Website Folder Concept

- A website consists of files and folders
- One folder acts as the root folder
- All web files exist inside this root folder
- The browser uses folder structure to locate files

Root Folder

- The root folder is the main folder of a website
- Represented by /
- All absolute (root-based) paths start from here



HTML File Paths

What is a path?

- A path tells the browser where a file is located
- Used in:
  - href (links)
  - src (images, media)
  - action
- File paths depend on the website folder structure
- 2 types of file paths
  - Absolute paths
  - Relative paths

Absolute Path - External URL

- Specifies the full location to an external resource
- Includes protocol (http or https)
- Does not depend on website structure

```
<a href="https://www.google.com">Google</a>
```

Absolute Path – Root-Relative Path

- Absolute within the website
- Starts from the root folder
- Begins with /

```
<a href="/pages/contact.html">Contact</a>
<img src="/images/logo.png">
```

Relative Path

- Location is relative to the current file
- Does not start with /

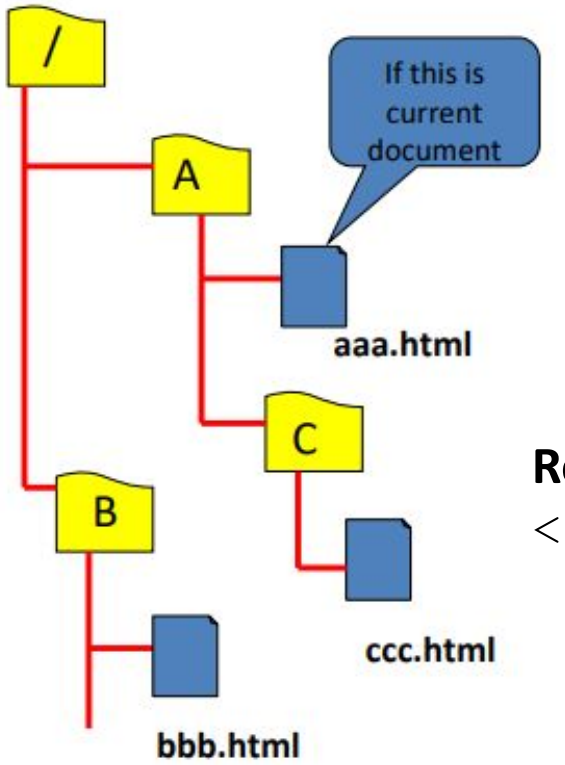
```
<a href="pages/contact.html">Contact</a>
<img src="images/logo.png">
```

Special Symbols in Relative Paths

. → current folder

.. → parent folder (one level up)

```
<img src="./images/logo.png">
<img src="../images/logo.png">
```



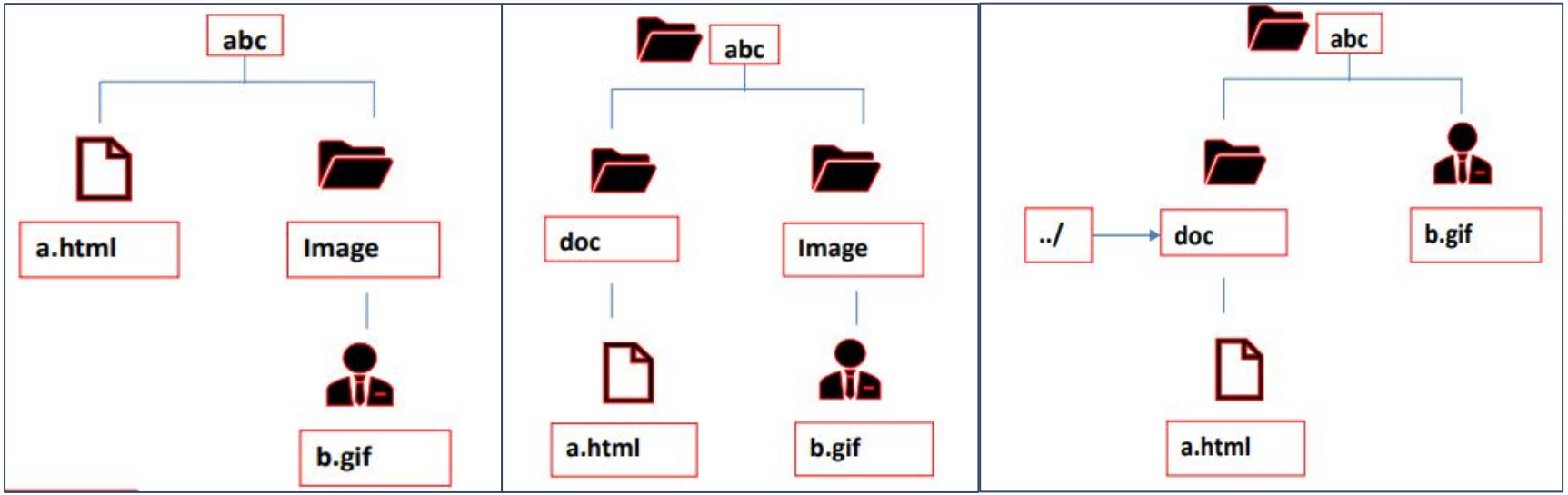
Absolute Path

```
<a href="/B/bbb.html">Go to bbb</a>
```

Relative Path

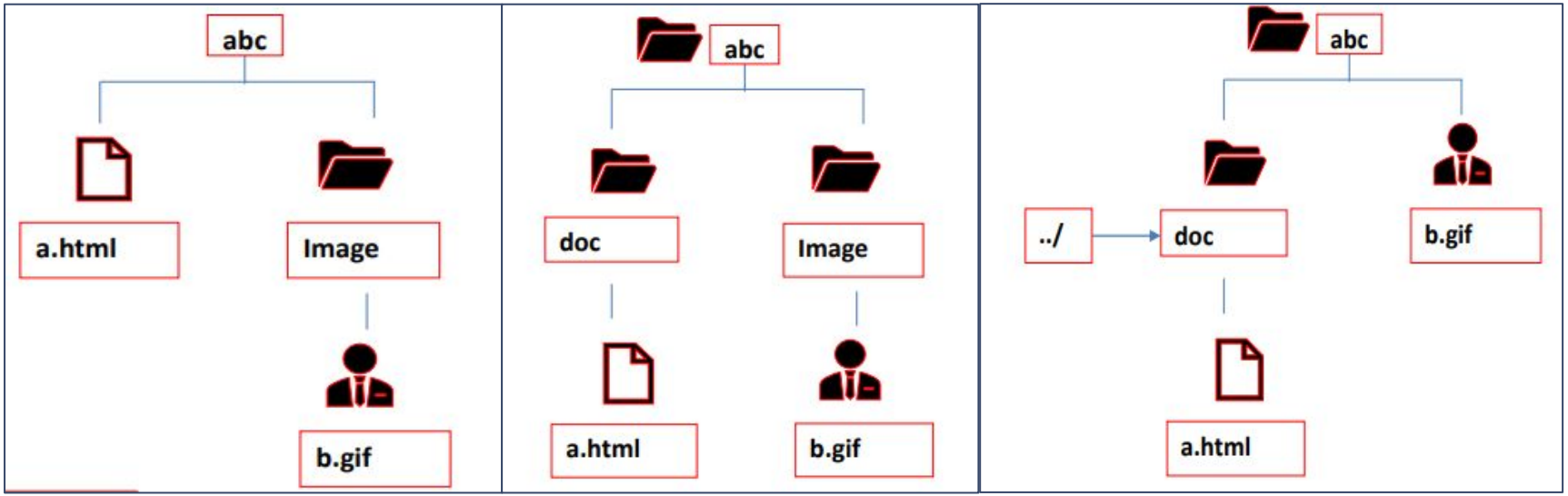
```
<a href="../B/bbb.html">Go to bbb</a>
```

Absolute Path and Relative Path



Identify the correct **absolute** and **relative paths** for the image file b.gif from the location of a.html in each diagram.

Absolute Path and Relative Path



<img src="/abc/Image/b.gif">
<img src="./Image/b.gif">
<img src="Image/b.gif">

<img src="/abc/Image/b.gif">
<img src="../Image/b.gif">

<img src="/abc/b.gif">
<img src="../b.gif">

What is a Bookmark in HTML?

- A bookmark allows users to jump to a specific location in a web page
- Also called an anchor
- Improves navigation in long pages
- Allows direct access to a specific section
- Reduces scrolling
- Created using:
  - id (target)
  - # (link)

Creating a Bookmark Target

```
<h2 id="contact">Contact</h2>
```

- id marks the destination
- id must be unique in the page

Linking to a Bookmark (Same Page)

```
<a href="#contact">Go to Contact</a>
```

- #contact links to id="contact"

Linking to a Bookmark in Another Page

```
<a href="about.html#team">Our Team</a>
```

- Opens about.html
- Scrolls to section id="team"

Image Mapping

- Image mapping allows **different areas of a single image** to act as separate hyperlinks
- One image → multiple clickable regions
- Uses three main components,
  - `<img>` with `usemap`
  - `<map>` with `name`
  - `<area>` to define clickable regions

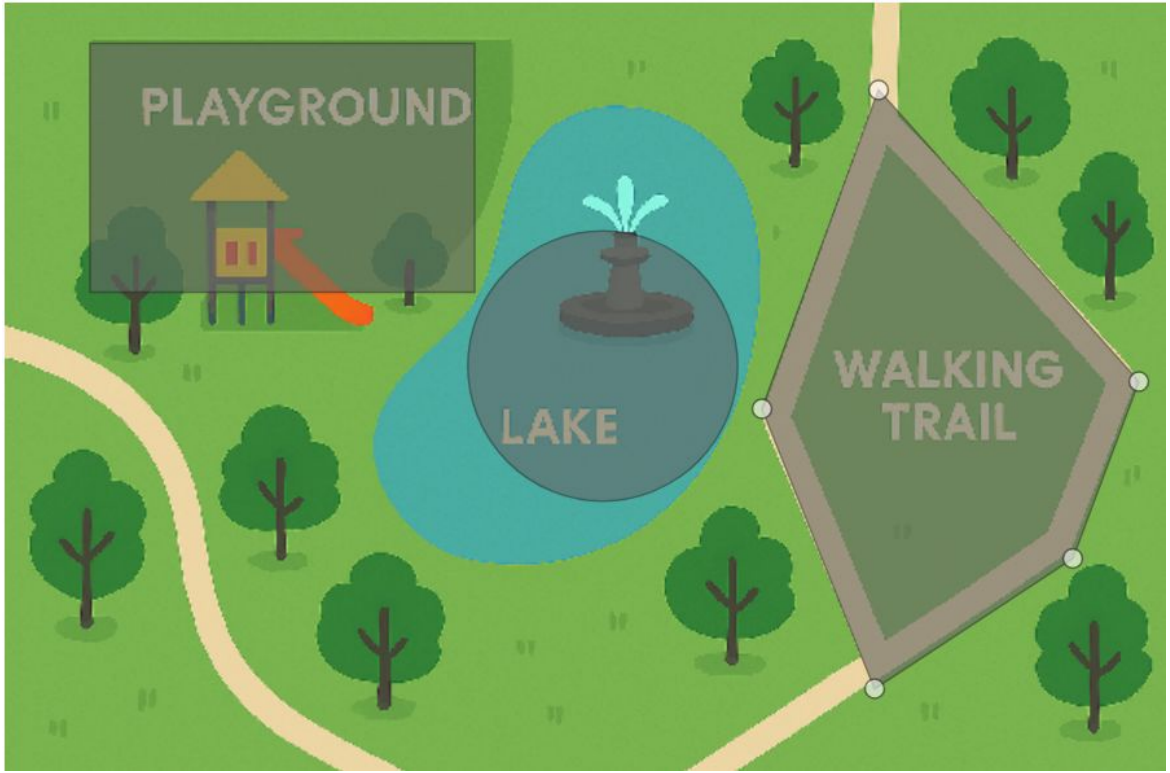
Mage Mapping

```
<img src="park.jpg" usemap="#parkmap" alt="City Park Map">
<map name="parkmap">
  <area shape="rect" coords="60,60,180,150"
href="playground.html" alt="Playground">

  <area shape="circle" coords="300,200,50"
href="fountain.html" alt="Fountain">

  <area shape="poly"
coords="400,100,450,120,480,180,420,200" href="trail.html"
alt="Walking Trail">
</map>
```

Mage Mapping

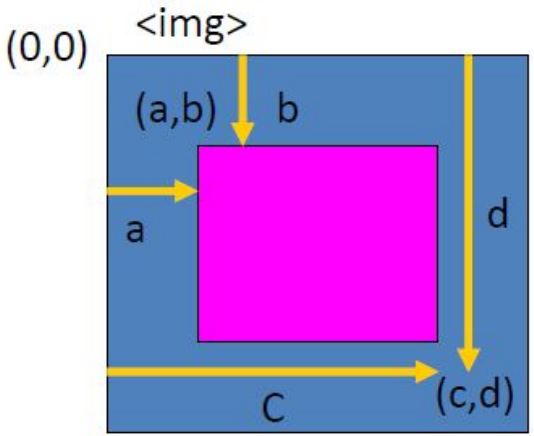


Shapes and Coordinates

- Coordinates are measured in pixels from the top-left corner of the image
- rect → rectangle
  - coords="x1,y1,x2,y2"
- circle → circle
  - coords="x,y,r"
- poly → polygon
  - coords="x1,y1,x2,y2,..."

Shapes and Coordinates

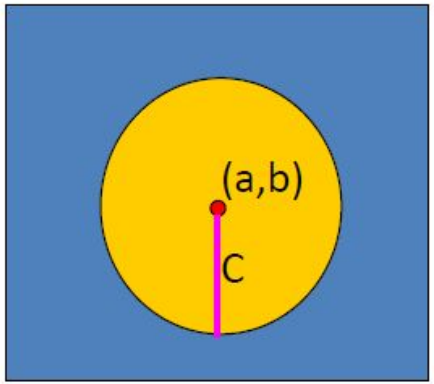
Shapes: RECT



Shape="rect"

Coords="a,b,c,d"

CIRCLE

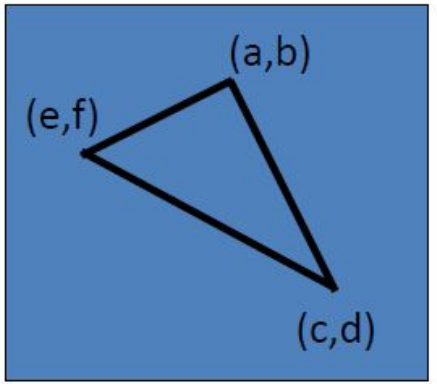


Shape="circle"

Coords="a,b,c"

(Centre & Radius –upto circumference of a circle)

POLY



Shape="poly"

Coords="a,b,c,d,e,f"

(coordinate of each corner point)

Thank You!